

Designing Precharge-Free Energy-Efficient Content-Addressable Memories

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Abstract—Content-addressable memory (CAM) is a specialized type of memory that facilitates massively parallel comparison of a search pattern against its entire content. State-of-the-art (SOTA) CAM solutions are either fast but power-hungry (NOR CAM) or slow while consuming less power (NAND CAM). These limitations stem from the dynamic precharge operation, leading to excessive power consumption in NOR CAMs and charge-sharing issues in NAND CAMs. In this work, we propose a precharge-free CAM (PCAM) class for energy-efficient applications. By avoiding precharge operation, PCAM consumes less energy than a NAND CAM, while achieving search speed comparable to a NOR CAM. PCAM was designed using a 65-nm CMOS technology and comprehensively evaluated under extensive Monte Carlo (MC) simulations while taking into account layout parasitics. When benchmarked against conventional NAND CAM, PCAM demonstrates improved search run time (reduced by more than 30%) and 15% less search energy. Moreover, PCAM can cut energy consumption by more than 75% when compared to conventional NOR CAM. We further extend our analysis to the application level, functionally evaluating the CAM designs as a fully associative cache using a CPU simulator running various benchmark workloads. This analysis confirms that PCAMs represent an optimal energy-performance design choice for associative memories and their broad spectrum of applications.

Index Terms—Content-addressable memory (CAM), fully associative cache, low-power, NAND CAM, NOR CAM, precharge-free.

I. INTRODUCTION

CONTENT-ADDRESSABLE memory (CAM) is an associative memory that provides fast, massively parallel search operations [1], [2]. This capability allows CAM to excel

in applications with intensive search workloads [3], [4]. While traditionally implemented with conventional embedded memory technologies [5], [6], [7], recent works have proposed CAMs based on emerging nonvolatile memory technologies [e.g., resistive RAM (RRAM), magnetic RAM (MRAM), 3-D-NAND, and ferroelectric field-effect transistor (FeFET)] [8], [9], [10], [11]. CAM designs, available in both binary and ternary variants, have found applications in various fields [1], [12], [13], [14]. While ternary CAMs (TCAMs) are particularly attractive for networking tasks, binary CAMs (BCAMs) are widely used in diverse domains such as computer architecture [15], [16], signal processing [17], data analytics [18], [19], and machine learning [20], [21]. Recently, CAMs have been used in applications involving approximate search operations, including big data analytics, machine learning, and DNA sequencing [19], [22], [23]. Unfortunately, CAM search capability comes at a higher power cost than traditional random access memory [1].

Standard CAM designs are classified into two types: NOR CAM and NAND CAM. Each type presents design trade-offs when implementing associative memories with regard to speed, power consumption, and scalability. Both solutions carry out search operations dynamically; starting with a precharge phase, followed by an evaluation phase that determines the search result. NOR CAMs are preferable in applications that demand high speed, but they can be excessively power-hungry [24]. On the other hand, NAND CAMs assure lower power consumption at the expense of reduced performance and scalability [1].

In this work, we introduce a novel class of precharge-free CAM (PCAM) architecture that provides a scalable, energy-efficient solution for the integration of associative memories in modern technologies. PCAM was implemented in a 65-nm CMOS process and evaluated through comprehensive circuit-level analysis, taking into account process variations and postlayout parasitics. Our analysis extends to the application level, where PCAM-based fully associative cache is benchmarked against conventional CAM and SRAM options using a CPU simulator running various workloads. When evaluated as a fully associative cache, PCAM achieves a $3.3\times$ reduction in energy consumption when compared to NOR CAM, while maintaining comparable search run times.

The key contributions of this work are listed below.

- 1) A novel PCAM cell is proposed, combining and leveraging the advantages of NOR and NAND CAMs.
- 2) PCAM is comprehensively evaluated under extensive Monte Carlo (MC) and process corner simulations while taking into account layout parasitics.

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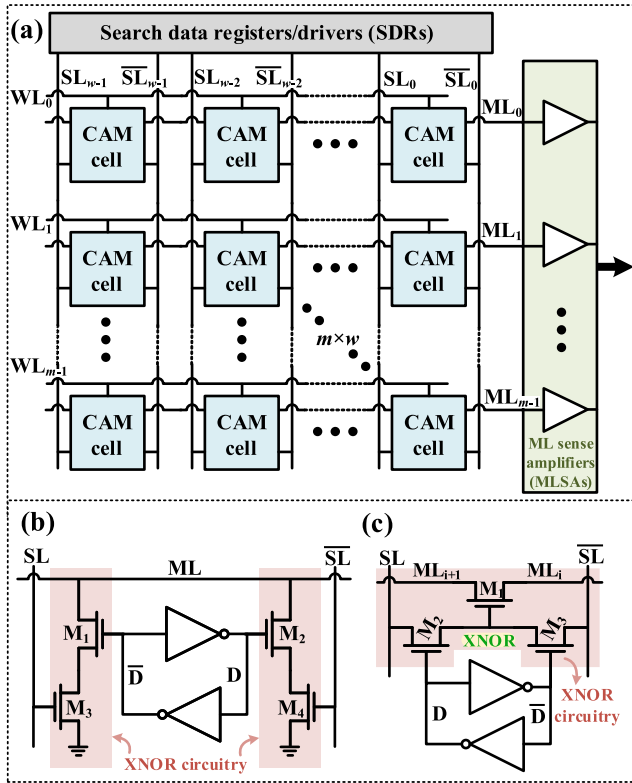


Fig. 1. CAM: (a) top-level view, (b) NOR CAM cell, and (c) NAND CAM cell. Note: access transistors and word lines are not shown in both NOR and NAND cells.

- 3) The precharge-free operation of PCAM is demonstrated to mitigate the limitations of conventional associative memory cells, such as the high energy consumption of NOR CAM and the limited number of bits in a segmented match line (ML) of NAND CAM.
- 4) PCAM is benchmarked against conventional CAMs, showing its effectiveness when used as a fully associative cache.

The rest of this article is structured as follows: Section II presents the background of conventional, low-power, and PCAMs; Section III details the proposed PCAM design. Section IV describes the evaluation and main circuit-level and application-level results. Finally, Section V presents the main conclusions of this work.

II. BACKGROUND

A. Conventional CAMs

Fig. 1(a) shows a CAM architecture organized as an $m \times w$ array of storage elements (CAM cells), where m represents the number of rows and w denotes the number of columns. This memory structure not only enables conventional read and write operations but also facilitates the comparison of a search pattern with the entire stored dataset. The generic row within the CAM array is linked to a dedicated ML, denoted as ML_i , where i ranges from 0 to $m-1$. For each row, the ML feeds the ML sense amplifier (MLSA) to signal whether the search pattern matches or mismatches the data word stored in the row. The search pattern held in the search data register (SDR) is driven onto the searchlines (SLs), SL_j and $\overline{SL}_j$ (with $j = w-1:0$), and compared in parallel with each

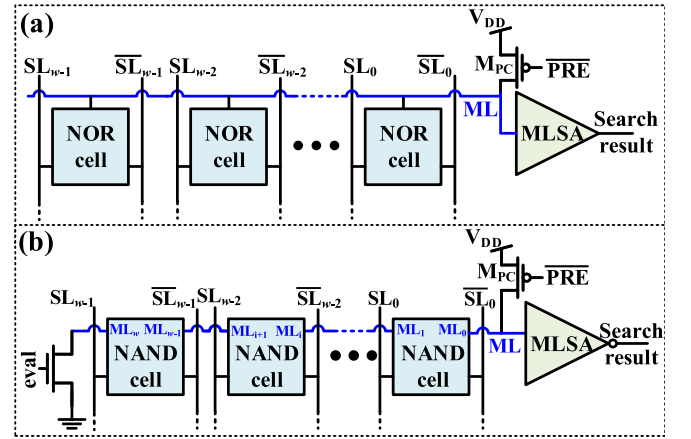


Fig. 2. CAM word typologies. (a) NOR CAM word and (b) NAND CAM word. Note: WLs are not shown.

memory row. Fig. 1(b) and (c) shows the two typologies of CAM bitcells: NOR CAM and NAND CAM [1]. The basic cell is based on the conventional six-transistor SRAM (6T-SRAM) bitcell, using cross-coupled inverters for data storage and pass-transistors connected to wordlines (WLs) for read and write access [not shown in Fig. 1(b) and (c)]. The CAM cell arrangement is such that every column is connected to a pair of SLs, while the ML spans across all cells in a row.

Fig. 2 focuses on the standard row configurations of NOR and NAND CAM. In NOR CAM [Fig. 2(a)], the cells in a row are connected in parallel, sharing the ML. On the other hand, in NAND CAM [Fig. 2(b)], the cells are serially connected to the ML. CAMs perform a comparison operation through two distinct phases: *precharge* and *evaluation*. In the precharge phase, SL and $\overline{SL}$ are both precharged to the ground, disconnecting all the ML pull-down paths (i.e., M_3 and M_4 in the NOR CAM cell and M_1 in NAND CAM cell are initially turned off). Meanwhile, the $\overline{PRE}$ signal, which controls the M_{PC} transistor in Fig. 2, is asserted to precharge the ML to V_{DD} . During the evaluation phase, the *inverted* search pattern is loaded onto the SLs, which are compared against the internal data nodes (D and $\overline{D}$) of the cells to identify matches or mismatches. In the NOR CAM design, when any stored bit equals the *inverted* query bit, the ML discharges, indicating a mismatch. Conversely, if every stored bit mismatches the respective *inverted* query bit, the ML remains high, signaling a match. Due to the short ML discharge path (limited to a couple of transistors), NOR CAMs are well-suited for high-speed search operations.

In the NAND CAM design, the nominal (direct) query pattern is asserted. If all bits stored in a row equal the query pattern, the ML discharges through a series-connected transistor chain [refer to transistor M_1 in Fig. 1(c)], thus signaling a match. However, if even just a single NAND cell mismatches the query pattern, there is no discharge path, and the ML remains high, indicating a mismatch.

In most applications, for example, cache tag arrays, at most one row will match during a search operation. For a NOR CAM, this means that all rows but one discharge during evaluation (or all rows, if the search pattern is not found in the CAM). Consequently, the following precharge cycle will need to fully charge the ML capacitance of $m-1$ (or m)

rows, resulting in very high power consumption. On the other hand, in NAND CAM, mismatching rows stay charged during evaluation, such that at the most, a single row needs to be charged during the next precharge cycle. While this leads to significantly lower power consumption than in a NOR CAM, the speed penalty for discharging a matching row through w series-connected nMOS transistors makes NAND CAMs considerably slower.

Another drawback of NAND CAM is the issue of charge-sharing between the ML node and the intermediate ML_i nodes. For example, in a NAND CAM row, a search pattern may match all the cells except for the leftmost cell in Fig. 2(b). In this case, charge sharing occurs between the ML and the intermediate ML_i nodes. During the evaluation, this may deteriorate the output level of the ML below the threshold voltage of the MLSA, thus erroneously signaling a match. The charge-sharing issue limits the number of cells allowed in a NAND CAM row, potentially requiring additional techniques at the expense of increased energy consumption and circuit complexity [1], [25].

It is worth noting that during the precharge phase, NOR and NAND CAMs unnecessarily toggle nodes with significant load capacitance, such as ML and SL, thereby wasting precious power. The power consumption of SL depends on its capacitance, which mainly includes the wire capacitance and one transistor gate (diffusion) capacitance per bit per column for the NOR (NAND) CAM. During the predischARGE of SL to the ground and the subsequent loading of search data onto SL, that is, transitions from $0 \rightarrow 0$ and $0 \rightarrow 1$, each SL transition contributes to the overall energy consumption per operation. As transistor dimensions shrink, wiring capacitance increases relative to the transistor parasitic capacitance, thus exacerbating power consumption during the precharge phase, particularly for advanced technology nodes [26].

B. Low-Power CAMs

Designing low-power BCAMs that support many bits per word while maintaining high-speed search capability is challenging. Jarollahi et al. [27] proposed an algorithm/architecture co-design that leverages sparse clustered networks to enable low-power operations. Miyatake et al. [28] presented a power-aware optimization strategy for CAMs tailored for application-specific domains. Ruan et al. [29] recommended implementing a precomputation stage to avoid the need for unnecessary precharging of all MLs. This technique was exploited in subsequent studies [25], [30], [31]. In particular, selective precharging of MLs through precomputation is presented in [31]. Hussain et al. [32] introduced a shared ML scheme, which triggers an ML discharge for every 2-bit mismatch, thus reducing the dynamic power consumption. Note that most of the reported works are based on NOR architectures, NAND solutions, or a combination of both, and they predominantly suffer from short-circuit current issues during the precharge phase or the aforementioned charge-sharing drawbacks during evaluation [1], [33].

Zackriya and Kittur [33] introduced an 8T CAM cell incorporating additional control circuitry per bit to avoid the precharge phase in each search cycle. However, for the proper functioning of the design, it necessitates a control signal to discharge the ML and internal nodes to avoid potential

false matches. Another work presents a self-controlled high-performance PCAM (SCPF-CAM) for high search frequency applications, ensuring faster data retrieval at the expense of high power consumption [34]. This design suffers from considerable short-circuit current due to forward-biased diode structures formed during ML evaluation [24].

Further low-power TCAMs have also been reported [35], [36], [37], [38], [39]. Ali et al. [35] provided a comprehensive review of low-power CAMs, highlighting different ML sensing schemes proposed in the 2000s, including precharge-high low-swing [40], selective precharge [41], pipelining [42], current-race (CR) designs [43], [44], and SL energy reduction techniques [45]. These approaches are mainly based on NOR-type ML schemes and present high design complexity or dedicated design optimization, limiting their power savings [35].

C. Segmented CAMs

Row segmentation is an appealing technique to enhance the power efficiency and performance of CAMs. While the segmented design methodology has been predominantly applied to NAND-based CAMs [46], [47], [48], [49], there have also been proposals for segmented NOR CAMs [2]. In this approach, the rows of the CAM array are divided into smaller segments, each featuring a local ML. All local MLs are then used as inputs to a NOR/OR logic tree, whose output is then sensed by a MLSA.

By segmenting the rows, the critical path of NAND CAMs can be shortened, since the discharge path has to traverse fewer transistors connected in series. In addition, row segmentation can mitigate the charge-sharing issue of NAND CAM structures, since the worst case ratio between the output capacitance and internal nodes can be limited to ensure proper functionality. Unfortunately, the segmentation of rows means that the power-saving features of NAND structures are reduced, since several segments may discharge in mismatched rows, requiring precharging on the following cycle. Therefore, the efficiency of applying row segmentation is highly dependent on the power performance and area-occupancy tradeoff created by the amount of segmentation.

As detailed in Section III, row segmentation is another major feature of the proposed PCAM design.

D. SOTA Summary

While previous CAM designs have explored techniques such as selective precharging and shared MLs to reduce power, they still fundamentally rely on the energy-inefficient precharge operation. To address this, in Section III, we introduce PCAM, a novel class of CAM cells that eliminates the need for precharge operation. Our precharge-free designs do not require additional circuitry that can lead to slower search speed, search errors, and short-circuit current issues. This is possible through a static precharge-free cell design that combines the advantages of both NOR and NAND CAMs. By avoiding dynamic precharging, PCAM offers a scalable and energy-efficient solution for CAM applications.

It is worth emphasizing that this work primarily focuses on introducing novel PCAM cell architectures, rather than exploring matchline sensing schemes.

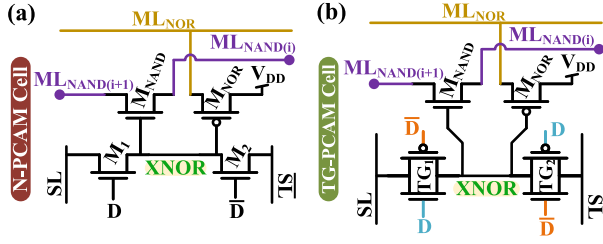


Fig. 3. Proposed precharge-free (a) N-PCAM and (b) TG-PCAM cells.

III. PCAM DESIGN

A. PCAM Cell

NMOS-based PCAM (N-PCAM)¹ and transmission gate PCAM (TG-PCAM) cells are presented in Fig. 3(a) and (b), respectively. As in conventional CAM cells, the storage block of both designs consists of a 6T-SRAM (not shown in Fig. 3). The comparison logic combines NOR and NAND CAM schemes, featuring M_{NOR} and M_{NAND} transistors connected to the ML_{NOR} and ML_{NAND} lines, respectively. The storage nodes, D and $\bar{D}$, of N-PCAM and TG-PCAM cells are connected to the gates of a pair of nMOS (M_1 and M_2) and complementary transmission gates (TG_1 and TG_2), respectively. The right and left diffusions of these devices are connected to SL and $\bar{\text{SL}}$, while sharing a terminal that forms the XNOR node, representing the comparison result (bitwise XNOR) between the stored bit and the query bit loaded onto the SL s. The XNOR nodes of both alternatives are connected to the gate terminal of the complementary pair of transistors, M_{NAND} and M_{NOR} . The M_{NAND} transistor is connected in series to the M_{NAND} transistors of the adjacent cells within the same row, similar to the NAND CAM configuration. The source of the M_{NAND} of the leftmost cell in a row is connected to the ground, which results in the discharge of the ML_{NAND} line when every XNOR node in the row is high. Due to the complementary functionality of the designs, neither the N-PCAM nor the TG-PCAM suffers from the charge-sharing limitation of the NAND CAM.

As for the M_{NOR} connection, one end is connected to V_{DD} , and the other to an ML_{NOR} line that is shared by all the cells in a row. This can be referred to as a segment of a PCAM row, as detailed in Section III-C.

B. Working Principle

Fig. 4 shows the timing diagrams of all the considered CAM types: NOR, NAND, N-PCAM, and TG-PCAM. The diagrams capture two consecutive search operations with different query patterns; the first yielding a match and the second, a mismatch. The conventional CAM designs operate in two phases, which are highlighted in Fig. 4: precharge, during which the ML is precharged to V_{DD} , and evaluation, where the ML is eventually discharged. For the NAND CAM and static designs, the XNOR nodes are also shown. In all designs, the ML voltage change is sensed by a MLSA. In dynamic NOR (NAND) CAM, during a match scenario, where the inverted (direct) query matches the data D , the ML remains (is discharged) high (down to ground). In a mismatch case, where

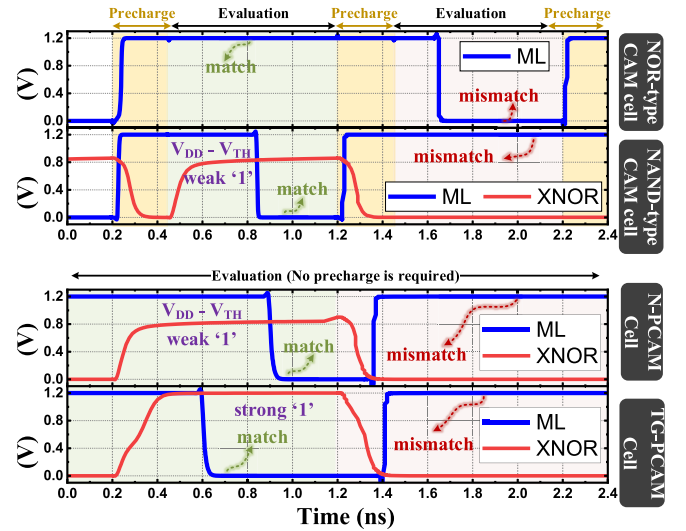


Fig. 4. Timing diagram of the ML and XNOR node for two consecutive search operations: match and mismatch.

the inverted (direct) query differs from D , the ML undergoes a complete discharge for the NOR CAM, while it is held high for the NAND CAM. For the N-PCAM and TG-PCAM designs, similar to a NAND CAM, ML is discharged down to the ground during a match and maintained or charged to V_{DD} during a mismatch.

Note that the voltage level at the XNOR node (labeled in Figs. 1(c) and 3) impacts the ML evaluation, that is, the ML charge or discharge rate. The XNOR node of both the N-PCAM and the NAND CAM is charged to $V_{\text{DD}} - V_{\text{TH}}$ (weak "1"). In contrast, the XNOR node of the TG-PCAM is fully charged to V_{DD} (strong "1"), due to the inclusion of transmission gates between the XNOR node and SL s.

C. PCAM Architecture

The PCAM precharge-free content addressable memory is built from PCAM cells of Fig. 3 and applies row segmentation to boost the performance. As shown in Fig. 5(a), a PCAM macro comprises: 1) $m \times w$ PCAM cells; 2) SDRs; and 3) an OR logic tree as part of a matchline sensing circuitry. Each w -bit PCAM row is partitioned into s distinct segments to limit the number of serially connected PCAM cells in the discharge paths. Such segmentation results in each pull-down path being reduced to w/s transistors connected in series. The outputs from all segments in a row are local MLs propagated through an OR logic tree with s inputs. Note that the output of the matchline sensing circuitry is represented by ML_i , that is, the output of the OR logic tree.

The transistor-level views of a segmented PCAM row implemented with the two PCAM cell variants, N-PCAM and TG-PCAM cells, are shown in the simplified schemes in Fig. 5(b) and (c), respectively. Within each segment, the pull-down (ML_{NAND}) and pull-up (ML_{NOR}) pathways are connected to the local ML. Fig. 5(a) shows both match (green) and mismatch (red) scenarios, specifically referring to the leftmost and right-most segments. In the context of a memory array, there is typically one match and $m - 1$ mismatch rows. The main source of energy consumption during the search operation is due to the mismatch in $m - 1$ rows.

¹This CAM cell was recently proposed in [50] to build a precharge-free very-many-way associative cache. This work reintroduces N-PCAM alongside the proposed TG-PCAM, establishing PCAM as a new class of CAMs.

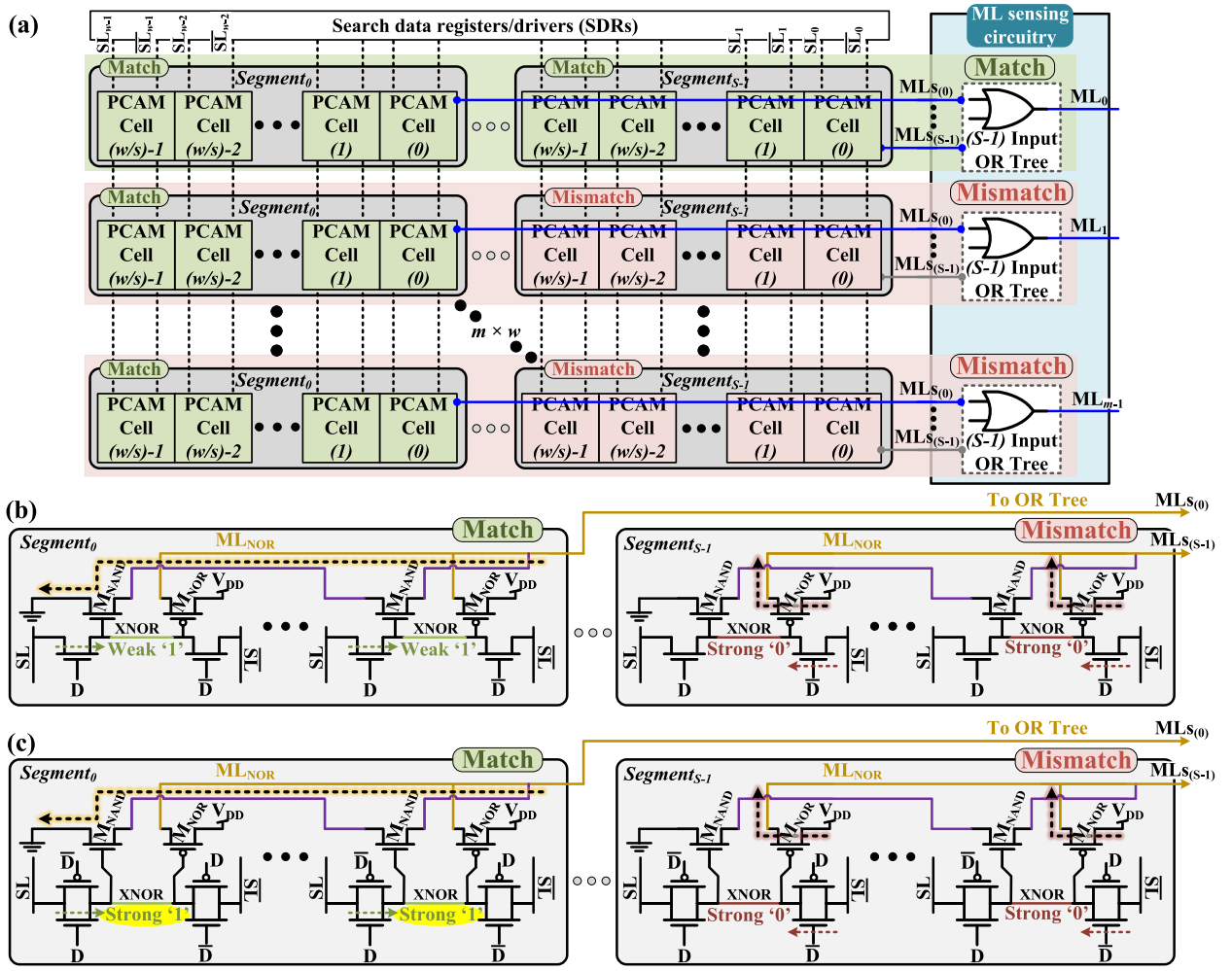


Fig. 5. PCAM architecture: (a) top-level structure, (b) N-PCAM simplified scheme, and (c) TG-PCAM simplified scheme. Note: SRAM cell is not shown. In the context of a memory array, it is important to note that there is one match and $m - 1$ mismatch rows. OR tree is part of the matchline sensing circuitry.

TABLE I

QUALITATIVE COMPARISON BETWEEN CONV. CAM AND PCAM DESIGNS

Design	Speed	Energy	Precharge	ML node	ML segmentation
NOR-CAM	high	high	Yes	Dynamic	Yes
NAND-CAM	low	low/mid	Yes	Dynamic	limited (charge sharing issues)
N-PCAM	low	low	No	Static	Yes
TG-PCAM	mid	low	No	Static	Yes

The search energy of a mismatch is affected by the number of segments matched (or discharged) in a row.

In the event of a mismatch, the XNOR node will present a strong “0,” charging the ML_{NOR} line (also referred to as the ML of a segment) to V_{DD} , and cutting-off (ML_{NAND} OFF) the segment discharge path. As for the match case, the XNOR node presents a weak “1” for the N-PCAM row, while a strong “1” is obtained at the XNOR node of the TG-PCAM, which uses TGs. This significantly improves the ML discharge time during a match. These energy contributions, as well as the search speed and area overhead, are evaluated hereafter.

Table I summarizes qualitative characteristics of conventional CAM typologies when compared to PCAM designs.

Due to ML architectural organization (see Fig. 2), NOR CAM provides high speed at the expense of power, whereas NAND CAM saves power at the expense of search speed. Both require energy-hungry precharge operation and NAND CAM are susceptible to charge sharing, limiting the number of bits in a segmented ML. The PCAM designs offer an energy-efficient solution for mid-low performance applications, thanks to their precharge-free operations without charge-sharing limitations.

IV. EVALUATION AND RESULTS

A. Methodology

The assessment of the proposed PCAM design was carried out in two phases: first, a circuit-level evaluation, followed by an application-level analysis, wherein PCAM was functionally evaluated as a fully associative cache. A 256×128 PCAM array was designed and laid out using a commercial 65-nm technology with a nominal voltage of 1.2 V. Circuit simulations were performed using Cadence Virtuoso under statistical MC simulations and process corner simulations. For reference, the discharge path of the ML was sized equally for all the cells (width of 200 nm). The remaining transistors were minimum-sized. All circuit-level analyses rely on postlayout simulations, fully accounting for both intercell and intracell RC parasitics. Moreover, the figures of merit reported in the circuit-level

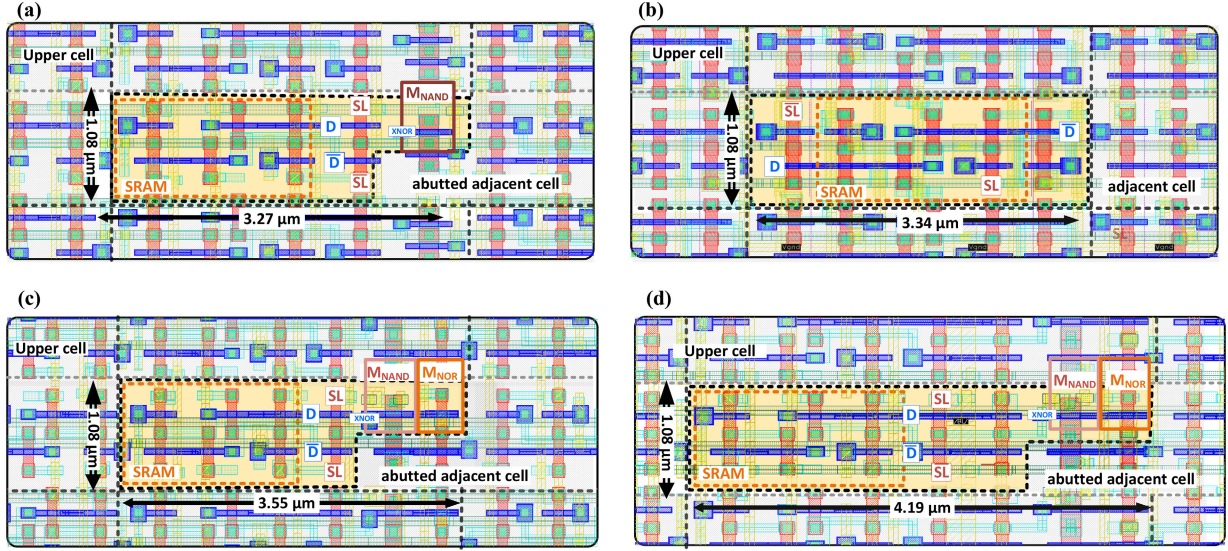


Fig. 6. Snippet of the 256×128 CAM and PCAM layout views, partly showing three rows and highlighting the core memory cell taking into account the intercell interactions: (a) NAND CAM, (b) NOR CAM, (c) N-PCAM, and (d) TG-PCAM.

results (Section IV-B) account for the impact of the matchline sensing circuitry (i.e., logic tree).

To assess the application-level performance, we used memory access traces generated by the Sniper CPU simulator [51] running the Splash benchmark [52]. The Sniper simulator was modified to account for the matched/mismatched segments of a segmented CAM. Throughout our analysis, the proposed TG-PCAM is benchmarked against its counterpart (N-PCAM) and the conventional dynamic NOR and NAND CAM counterparts.

B. Circuit-Level Results

1) *Cell Layout*: Fig. 6 shows the physical views of all CAM cells evaluated in our analysis. A similar layout strategy has been adopted for all the cells, featuring a $1.08\text{-}\mu\text{m}$ tall 6T SRAM with pitch-matched extra nMOS transistors placed on the sides. A dense array was obtained for Fig. 6(a) NAND CAM, (c) N-PCAM, and (d) TG-PCAM by abutting M_{NAND} and M_{NOR} transistors placed on the sides with the adjacent cell. Although the conventional Fig. 6(b) NOR CAM cell ($3.61\text{ }\mu\text{m}^2$) allows straightforward integration with the 6T SRAM, without shared diffusion with an adjacent cell it occupies 2% larger area compared to (a) NAND CAM cell ($3.53\text{ }\mu\text{m}^2$). Fig. 6(c) N-PCAM ($3.84\text{ }\mu\text{m}^2$) cell includes an additional pMOS transistor (M_{NOR}) compared to (a) NAND CAM cell. The additional devices located on the edge of the cell do not share the same N-WELL with the SRAM, thus leading to an 8.6% area penalty. Finally, Fig. 6(d) TG-PCAM ($4.53\text{ }\mu\text{m}^2$) uses two additional pMOS transistors as components of the transmission gates [refer to Fig. 3(b)] with an area increase of 26% in comparison to the NOR CAM cell.

Fig. 7 shows the breakdown of the area occupation for 256×128 memory arrays for different segmented configurations (2–32 segments). The logic tree area overhead increases with the number of segments mainly due to the increased depth of the logic tree circuitry and interconnection complexity. The logic tree area accounts for 1.39% (on average) to 24.8%

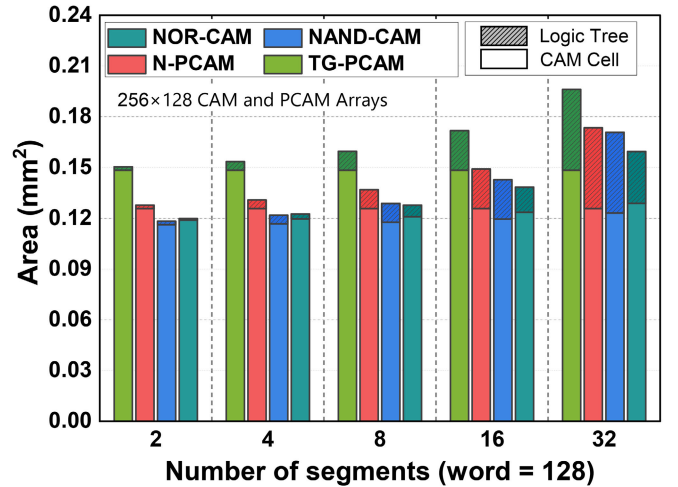


Fig. 7. Area footprint versus number of segments for a 256×128 CAM array.

(on average) of the total area of the CAM array as the number of segments increases from 2 to 32. The CAM cells occupy 98.6%, 96.5%, 92.7%, 85.9%, and 75.3% of the total area for 2, 4, 8, 16, and 32 segment designs, respectively. Note that logic tree circuitry of NAND-based designs inherently requires an extra inverter per segment, leading to the total area of NAND CAM surpassing that of NOR CAM as the number of segments increases.

Conventional NOR and NAND CAMs require one precharge transistor per segment, contributing up to 6% of the total area in 32-segment designs.

Note that as the number of segments increases, more metal layers are required to route the MLs within the CAM cell pitch. The $(S-1)$ -input logic tree implemented with two-input logic gates and buffers placed along the CAM word groups the MLs to alleviate wiring capacity and routing congestion. For a CAM cell pitch of $1.08\text{ }\mu\text{m}$, up to four horizontal routing tracks for metal-3 (M3) and metal-4 (M4) are allowed.

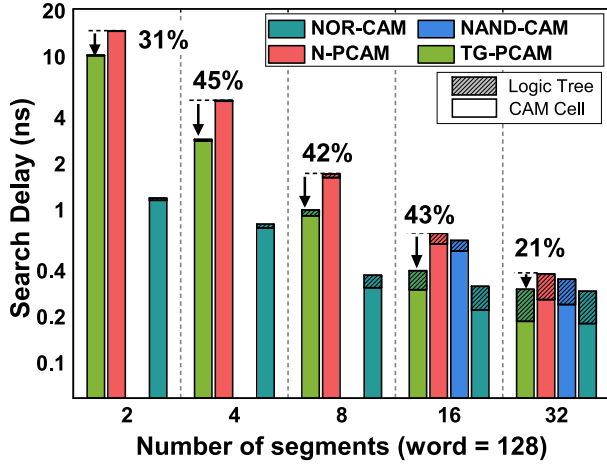


Fig. 8. Postlayout search delay versus number of segments for 256×128 CAM array.

In the 32-segment configuration, the most congested region required four M3 and three M4 routing tracks for all the MLs, utilizing a 32-input, five-level depth logic tree.

2) *Delay*: The search delay considered in our analysis includes the impact of all the circuitry presented in Fig. 5, including the matchline sensing circuitry, that is, logic tree. Fig. 8 shows the search delay results for conventional CAM and PCAM arrays, organized with 256 rows of 128-bits, divided into different numbers of segments (i.e., 2, 4, 8, 16, and 32). Increasing the number of segments results in a deeper logic tree, leading to a significant delay overhead. For 32-segment designs, this overhead can reach up to 38% of the total search delay, as shown by the stacked filled bars in Fig. 8). As the number of segments increases, the search delay decreases, mainly due to the reduced number of devices in the ML discharge path (connected to the ML) of the PCAMs/NAND CAM (NOR CAM). As expected, the NOR CAM is always the fastest option. For the 32-segment words, involving four M_{NAND} transistors in the ML discharge path, the NAND CAM and N-PCAM solutions present similar search delays. Notably, the TG-PCAM demonstrates a significant performance advantage when compared to both N-PCAM and conventional NAND CAM, leveraging the full swing at the XNOR node [refer to Fig. 5(c)]. It achieves an average delay reduction of 25% and 36% when compared to the NAND and N-PCAM cells, respectively. TG-PCAM achieves a sub-1-ns delay starting with eight segments per 128-bit word.

Note that the implementation of the NAND CAM has been limited to a minimum of 16 segments (i.e., a maximum of 8-bits per segment) due to the charge-sharing issue that can lead to false matches [1]. This limitation is effectively addressed by PCAM designs, which can be implemented with fewer segments per word, thus mitigating additional circuitry and complexity. Overall, the TG-PCAM design emerges as a performance-efficient alternative to NOR CAM, with only a 3% increase in the search delay for the case of a 32-segment configuration.

3) *Energy*: The energy consumption was evaluated under a realistic assumption that a single CAM row returns a match, while the remaining 255 rows mismatch. The energy results

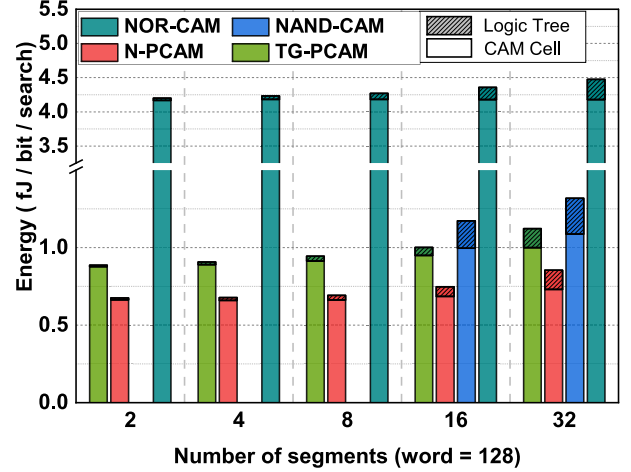


Fig. 9. Postlayout search energy per bit versus the number of segments for 256×128 CAM arrays. The results are referred to the case when 50% of segments of a 128-bit word are discharged during a mismatch case.

reported hereafter pertain to the predominant mismatch. We further assume that 50% of the segments of a 128-bit word are discharged, while the others remain charged throughout the search cycle.² The energy consumption is evaluated by postlayout simulations for all CAM configurations.

Fig. 9 shows the average search energy per bit, taking into account different segmentation (from 2 to 32 segments) of the CAM word. The intrinsic CAM cell energy consumption contribution of the logic tree is also considered (refer to stacked bars with a filled pattern in Fig. 9). As opposed to the CAM performance trend, observed in Fig. 8, the search energy increases as the number of segments within the word rises. This is mainly due to the increased depth of the logic tree circuitry, wherein an increased number of logic tree levels become active/toggling. The logic tree accounts for 9% (on average) and 29% of the total energy per bit per search in PCAMs and NAND-CAM designs, respectively. The NOR CAM presents an energy per bit of about 4.2 fJ for the two-segment design. As the number of segments within the word increases, the average bit energy increases as well, up to more than 4.4 fJ for the 32-segment configuration. As for the NAND, N-PCAM, and TG-PCAM arrays, the average energy consumption is about 1.23, 0.97, and 0.72 fJ, respectively. TG-PCAM presents higher energy consumption per bit per search than N-PCAM, primarily due to the parasitic capacitance associated with its XNOR circuitry (see Fig. 3).

Interestingly, while PCAM cells have larger parasitic capacitance, NAND CAM consumes more energy per bit per search. This is mainly due to the toggling of the SL wire and diffusion capacitances during the precharge phase (refer to the end of Section II-A).

4) *Process Variability*: We extended our analysis of CAM designs to evaluate the impact of process variability on delay and leakage current through MC simulations. The 16-segment configuration was considered for this analysis.

²Note that a lower (higher) percentage of segments discharged leads to lower (higher) search energy consumption per bit. The actual number of discharged segments is access pattern dependent, however the 50% assumption allows us to fairly compare both NOR and NAND type CAMs with a similar input pattern.

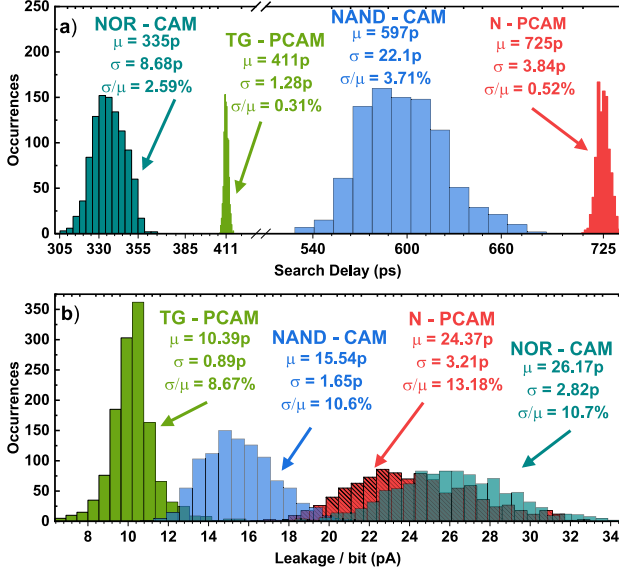


Fig. 10. MC simulation results in terms of (a) search delay and (b) leakage for TT process corner. Results are based on 1 K MC samples for a 16-segment 256×128 CAM.

MC simulation results for the typical-typical (TT) corner are shown in Fig. 10. The NOR CAM is the fastest solution, while the TG-PCAM reduces leakage power with the lowest variability ($\sigma/\mu = 8.7\%$). The N-PCAM cell presents about $1.56\times$ and $2.34\times$ higher leakage power than NAND CAM and TG-PCAM cells, respectively. This is attributed to the voltage level at the XNOR node (see Fig. 3), which leads the M_{NOR} pMOS transistor to have $V_{\text{SG}} = V_{\text{TH}}$ and $V_{\text{SG}} = 0$ for the N-PCAM and TG-PCAM designs, respectively.

Table II summarizes the corner analysis results for slow-slow (SS) and fast-fast (FF) process corners when compared to the TT case. TG-PCAM is shown to maintain its advantages in terms of both delay and leakage when compared to the NAND CAM and N-PCAM designs. The N-PCAM results in the less energy-hungry option over the different process corners.

5) *Voltage Scalability*: We performed MC simulations to assess the impact of voltage scaling on search delay. Fig. 11 shows the search delay as a function of the supply voltage. Considering a target search period of 8 ns, TG-PCAM achieves a V_{min} of about 0.7 V, which is 22% lower than N-PCAM V_{min} of 0.9 V. This improvement is mainly due to the full swing at the XNOR node. Unlike conventional NAND CAM, which has a V_{min} limited by charge-sharing issue, PCAMs and NOR CAM are operative even for V_{DD} down to 0.5 V. When compared to NAND (NOR) CAM, TG-PCAM is $4\times$ (85%) faster (slower) when operating at a V_{DD} of 0.7 V (0.5 V).

6) *Comparison With SOTA*: Table III summarizes post-layout simulation results of NAND, NOR, N-PCAM, and TG-PCAM designs and compares them to prior work. In addition, the current race (CR) CAM [1], [43] organized as 256 rows of 128-bits, was included in our comparison. This design avoids the need for ML precharging during compare operations. Among the different CAM configurations designed in this work, our focus for this comparison is mainly on 16-segment designs. The conventional CAMs, CR CAM, and

TABLE II
PROCESS CORNER ANALYSIS FOR A 16-SEGMENT 256×128 CAM

Process Corners	Search Delay (ps)			Energy / bit / search (fJ)			Leakage / bit (pA)		
	SS	TT	FF	SS	TT	FF	SS	TT	FF
NOR-CAM	486	323	186.5	3.98	4.36	4.92	21.2	26.2	92.6
NAND-CAM	1252	602	360	0.93	1.17	1.48	11.6	15.5	20.3
N-PCAM	1750	718	379	0.54	0.74	0.88	18.2	24.4	50.9
TG-PCAM	550	409	313	0.99	1.05	1.12	8.2	10.3	17.6

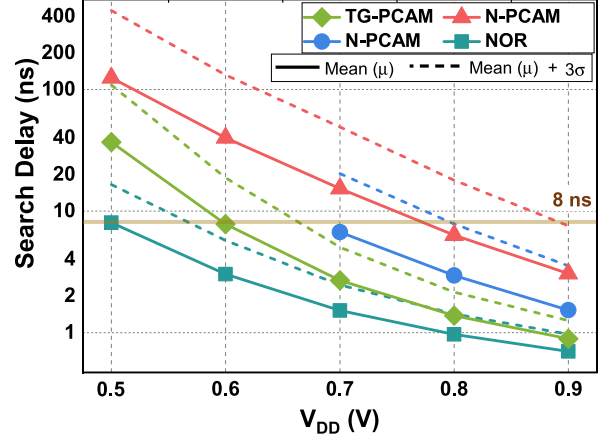


Fig. 11. Voltage scalability: MC simulation results in terms of search delay as a function of the supply voltage for 16-segment 256×128 PCAM and CAMs. Results are referred to the TT process corner.

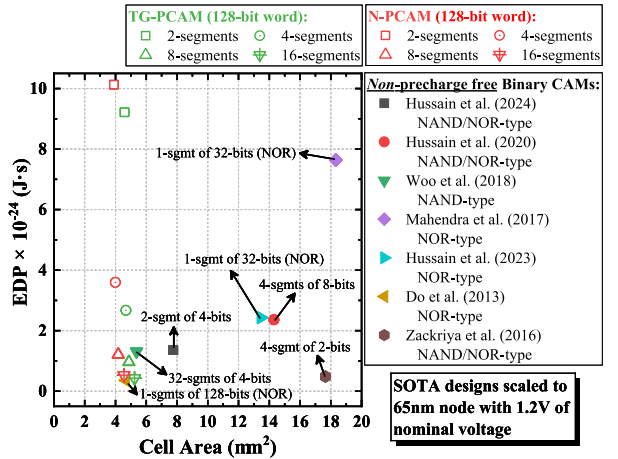


Fig. 12. EDP versus area comparison of SOTA CAM structures and PCAMs. SOTA designs are scaled to the 65-nm node [58]. SOTA: Hussain et al. [53], Hussain et al. [54], Woo and Yang [48], Magendra et al. [34], Hussain et al. [32], Do et al. [30], and Zackriya and Kittur [33]. Note: SOTA nonprecharge-free CAMs require precharge and evaluation cycles.

PCAM cell area footprint account for the overhead of logic tree circuitry.

Considering search speed and excluding the NOR CAM due to its notable energy consumption, TG-PCAM proves to be a very appealing solution, ensuring search operations that are 43% and 32% faster than N-PCAM and NAND architectures, respectively. In a broad context of state-of-the-art (SOTA)

TABLE III

FEATURES SUMMARY AND COMPARISON. THE FIGURES OF MERIT REPORTED IN THIS WORK ACCOUNT FOR THE INTRACELL AND INTERCELL WIRING OF BOTH THE CAM CELLS AND THE LOGIC TREE. SOTA NONPRECHARGE FREE CAMs REQUIRE PRECHARGE AND EVALUATION CYCLES

	[53]	[54]	[48]	[34]	[30]	[32]	[33]	This work					
Technology (nm)	45	45	65	45	65	45	45	65 (45) ^d					
Supply Voltage (V)	1.0	1.1	1.0	1.0	1.0	1.0	1.0 ^c	1.2 (1.0) ^d					
ML Structure	NAND/NOR	NAND/NOR	NAND	NOR	NOR	NOR	NAND/NOR	NAND	NAND	NAND	NOR	NOR	NOR
ML precharge	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	NO	NO
Approach/Scheme	Adaptive Selective ML	Pseudo-static master-slave	Don't care reduction scheme	Self controlled Precharge	Power gated ML sensing	Shared ML	Selective Precharge	TG-PCAM	N-PCAM	Conventional ^e		Current Race (CR) Scheme ^e	
Cell Type	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM	BCAM
CAM Array	64 × 32	64 × 32	256 × 128	128 × 32	8K × 128	64 × 32	8 × 8	256 × 128					
# of segments	2	4	32	1	1	1	4	16	16	16	16	16	2
# of bits per segment	4	8	4 ^b	32	128	32	2	8	8	8	8	8	64
Search delay (ns)	1.04	1.09	2.5	1.25	0.75	0.79	0.31	0.41 (0.28) ^d	0.72 (0.51) ^d	0.60 (0.46) ^d	0.32 (0.29) ^d	0.28 (0.19) ^d	0.83 (0.59) ^d
Energy (fJ/search/bit)	0.45	0.96	0.41	2.1	0.76	1.06	0.53	1.05 (0.40) ^d	0.74 (0.30) ^d	1.17 (0.48) ^d	4.36 (1.78) ^d	6.68 (2.73) ^d	3.52 (1.44) ^d
Cell Area (μm ²)	5.08	9.34	5.37 ^a	12	4.68 ^f	8.81 ^a	11.56	5.24 ^g (3.43) ^d	4.55 ^g (2.98) ^d	4.36 ^g (2.86) ^d	4.22 ^g (2.76) ^d	33.1 ^g (21.3) ^d	4.67 ^g (3.06) ^d
EDP·A (J·s·m ²) × 10 ⁻³⁷	23.8	97.7	55.0	315	27	73.8	19.0	22.6 (3.84) ^d	24.2 (4.56) ^d	30.6 (6.31) ^d	58.9 (14.2) ^d	137 (26) ^d	621 (110) ^d

^a Norm. to 1 bit area. ^b Cascaded 3-level tree. ^c Not reported (obtained from waveform). ^d Value scaled-down to 45 nm node. ^e CAMs/CR from [1], [43].

^f Estimated. ^g Cell area accounts for logic tree (all designs), precharge transistors (only NOR/NAND CAMs), and CR sensing circuitry (only CR CAM)

designs, TG-PCAM is 32% slower than the hybrid NAND/NOR circuit presented in [33], although their result is reported for a much smaller array (only 8 × 8 cells), designed in a denser (45 nm) technology. Among the considered NAND-based solutions, N-PCAM is demonstrated to be energy-efficient while assuring faster search operation for larger CAM arrays. Among the two-segment designs, that is, CR-based and conventional NOR-based CAMs, the CR scheme demonstrates superior speed and energy efficiency, albeit at the cost of a larger area footprint. However, when increasing the number of segments to improve search speed, both area and energy consumption per search operation are adversely affected. Specifically, a 16-segment CR-based CAM consumes approximately 53% more energy than its 16-segment NOR counterpart. Such energy overhead can be mainly attributed to the larger number of transistors required by the CR scheme.

Along with lower energy consumption, PCAM offers additional advantages over CR-based CAMs [38], [43], [55], [56]. PCAM exhibits lower design complexity/effort, could be integrated more easily into memory compilers due to its digital-based ML sensing, and scales better to advanced technology nodes, for example, 16-nm FinFET [50] and below. In contrast, CR CAMs, which rely on analog-based sensing, demand higher design effort, face scalability challenges at advanced nodes due to reduced sensing margins [38], and require a predischARGE phase before each compare operation.

We introduce a combined figure-of-merit that integrates the energy–delay product (EDP) with the area footprint, denoted as EDP·A [57]. This metric allows evaluation of the tradeoff between performance and physical implementation efficiency. The proposed PCAMs demonstrate significantly lower EDP·A values when compared to conventional NAND and NOR CAMs designed in this work. Moreover, PCAMs exhibit the lowest

EDP·A compared to SOTA BCAM designs operating at the same technology node.

Fig. 12 shows a comparative analysis of EDP as a function of cell area footprint for PCAMs and SOTA designs. This comparison is carried out under the same conditions, that is, at the 65-nm node and operating at 1.2 V. For the SOTA designs, we estimate figures-of-merit at 65 nm using the scaling equations introduced in [58], which have been used frequently to assess CAM designs [59], [60], [61]. This analysis reveals the following key findings.

- 1) Nand-based BCAMs reported in literature [32], [33], [34], [48], [53], [54] typically use a limited number of bits per segment due to the charge-sharing issues (refer to Section II-A). PCAM designs overcome this limitation, allowing a larger number of bits per segment in NAND schemes.
- 2) Although the cell area footprint of PCAM includes the logic tree circuitry, it offers a very compact design compared to other SOTA designs.
- 3) The 8-segment and 16-segment PCAM designs are the best choices in terms of EDP and area footprint.

Overall, PCAMs achieve competitive performance and energy consumption figures-of-merit, while having relatively low cell area. The choice between N-PCAM and TG-PCAM solutions involves a tradeoff between delay and energy, where N-PCAM is preferable in low-power settings while TG-PCAM achieves high performance and best energy efficiency.

C. Application-Level Results

For system-level evaluation, both conventional CAMs and PCAMs were operated as fully associative caches, using memory traces generated by the Sniper CPU simulator running Splash benchmark workloads *barnes*, *cholesky*, *fft*, *radix*,

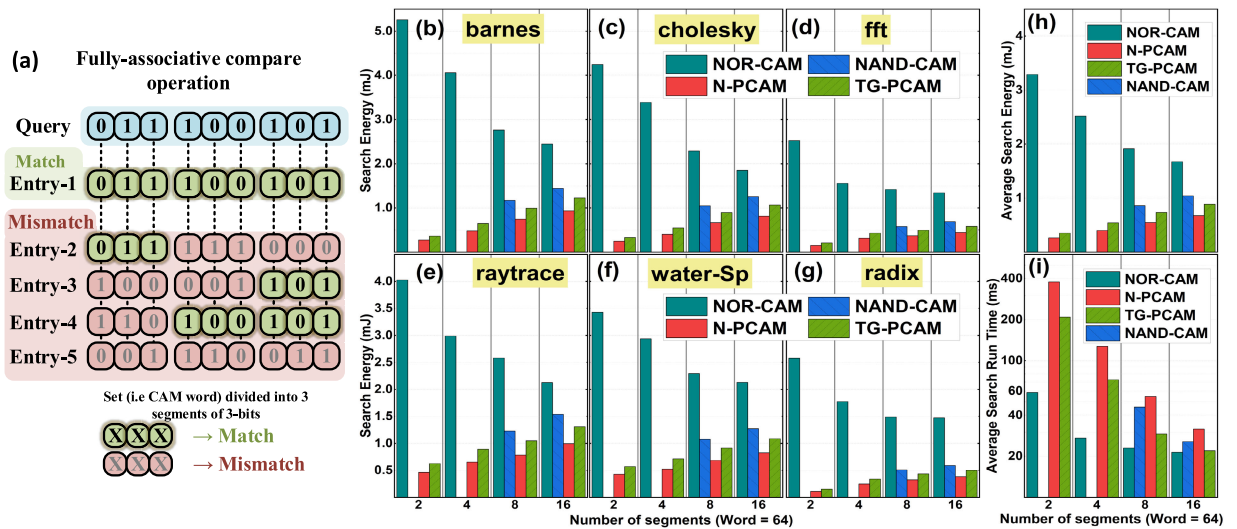


Fig. 13. (a) Illustration of a fully associative compare operation. In this example, each set (i.e., CAM word) is divided into three segments of 3-bits. (b)–(i) Application-level simulation results in terms of search energy and run time as a function of the number of different segmented schemes for fully associative cache configurations based on conventional and proposed CAMs: (b)–(g) search energy for *barnes*, *cholesky*, *fft*, *raytrace*, *water-Sp*, and *radix* benchmarks, respectively; (h) and (i) search energy consumption and search run time averaged over the considered set of benchmarks.

raytrace, and *water-Sp* [51], [52]. For this study, the fully associative cache size is 16 kB while the tag size is 64 bits.

In each run, for each access to the fully associative cache, we have extracted the number of segments that match and mismatch between the query tag and stored tags. This is illustrated in the example of Fig. 13(a): a query tag compared to a cache with five entries, each of which is divided into three segments of three bits. During search operations, the energy consumption mainly depends on the number of segments discharged, that is, mismatched (matched) segments in the case of a NOR CAM (NAND CAMs and PCAMs).

Fig. 13 shows the outcomes of application-level simulations in terms of total search energy consumption [Fig. 13(b)–(h)] and search run time [Fig. 13(i)] for different segmentation schemes (2, 4, 8, and 16). As expected, the energy consumption of PCAMs and NAND CAM reduces as the number of segments within the word decreases. This is because longer segments are less likely to match and discharge. While NOR-CAM is also segmented for a fair comparison, it stands out as the most power-hungry design for all segmented configurations. Different from the other designs, a smaller number of segments in NOR-CAM leads to higher energy consumption because longer segments are more likely to mismatch and discharge MLs.

Fig. 13(h) and (i) show the search energy and runtime averaged for all the considered benchmarks. The speed advantage of the NOR CAM is reduced on average as the number of segments within the word increases. Specifically, for the 16-segment case, the TG-PCAM consumes much less energy consumption than a NOR CAM, while maintaining comparable search speed. The latter presents 36% lower run-time compared to NAND CAM for the 16-segment configuration. While N-PCAM is the slower solution, it represents the less energy-hungry option. N-PCAM and TG-PCAM always demonstrate, on average, lower average search energy by 55% and 17%, respectively, when compared to NAND CAM. For the 8-segment designs, TG-PCAM emerges as the optimal choice.

When compared to the NOR CAM, TG-PCAM presents a delay penalty (21%) and it shows $2.62\times$ better energy consumption.

V. CONCLUSION

This work presented PCAM, a novel class of precharge-free content-addressable memories for high-performance, low-power applications. The proposed TG-PCAM design was shown to provide search speeds similar to NOR CAM with the low-power operation of NAND CAM. The N-PCAM option provides even lower power consumption with intermediate performance. System-level results, obtained for a set of different benchmark applications, demonstrate that PCAMs achieve considerable energy savings when compared to conventional fully associative caches using NOR CAMs. This energy efficiency is obtained while ensuring comparable search runtimes in highly segmented designs. Our findings suggest that PCAMs are an energy-efficient and high-performance design choice for associative memories and their broad spectrum of applications.

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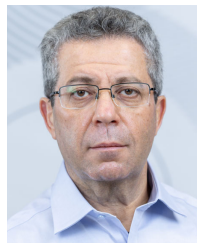


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